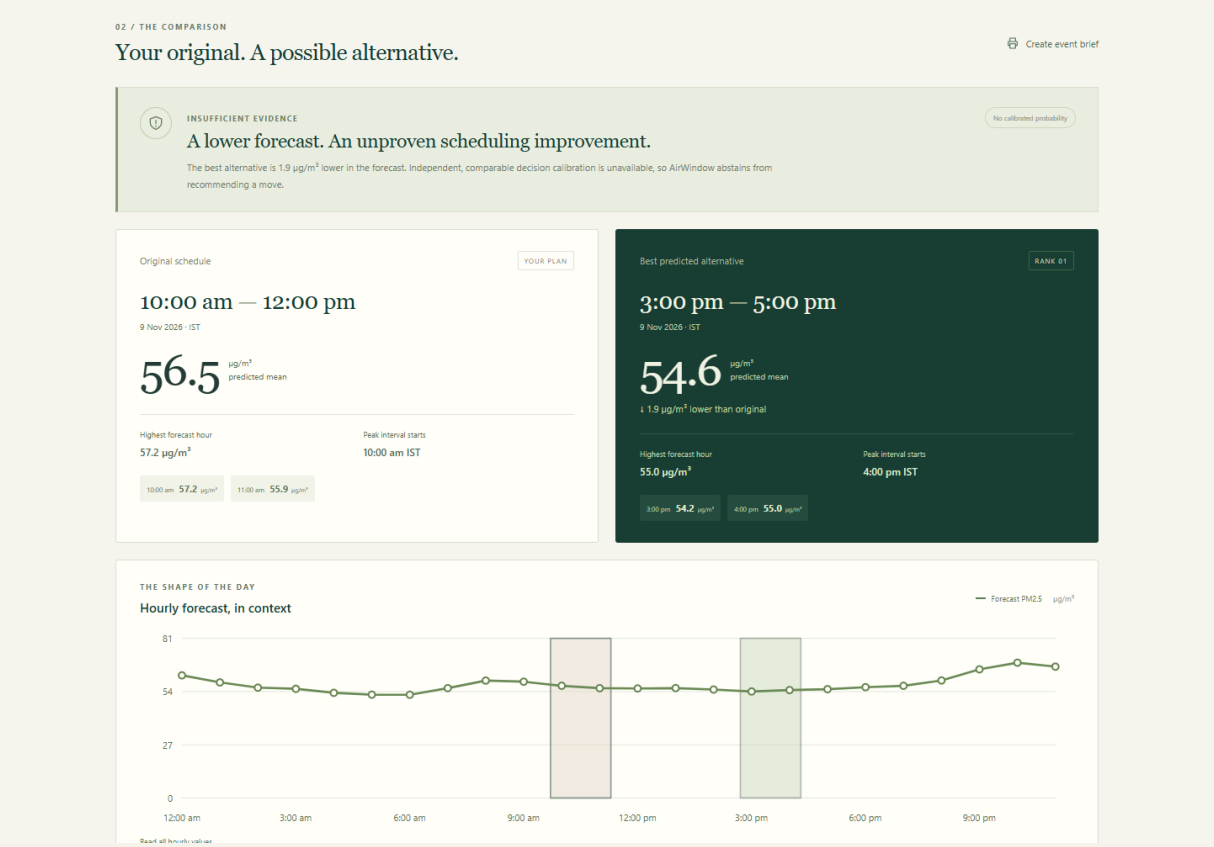
A PRACTICAL DECISION, WITH ITS LIMITS ATTACHED

Should you move your outdoor event?

AirWindow Mumbai

A confidence-aware activity scheduler for school sports coordinators and outdoor-event organisers.



Real application screenshot. Forecast values come from the participant model artifact; no synthetic production data.

Compare your original booking with complete alternatives. See the predicted difference, then see whether historical evidence is strong enough to act on it.

Current evidence is uncalibrated. A lower forecast is not a guarantee, and the product is allowed to abstain.

01 / THE PLANNING WORKFLOW

A booking is the starting point.

A coordinator has a two-hour event already scheduled. It can move between an earliest start and latest finish. AirWindow keeps that original visible while it compares the available options.

Configure	Choose one actual monitor, original start, duration 1/2/3h and feasible range.
Compare	Every candidate requires complete consecutive hourly forecasts. Show mean, maximum, peak hour and difference from the original.
Assess evidence	Show a prominent decision state, comparable support and validation limits. Do not turn numerical rank into certainty.
Replay	Inspect what a genuine historical forecast knew, its selected slot and subsequent actual observations.
Share	Print a one-page event brief with comparison, model version, timestamps and limitations.

Four states, with permission to abstain

ROBUST IMPROVEMENT is reserved for independently validated decision reliability. **TENTATIVE IMPROVEMENT** identifies an unproven numerical alternative. **NO CLEAR WINNER** permits ties or indistinguishable options. **INSUFFICIENT EVIDENCE** makes missing comparable support explicit.

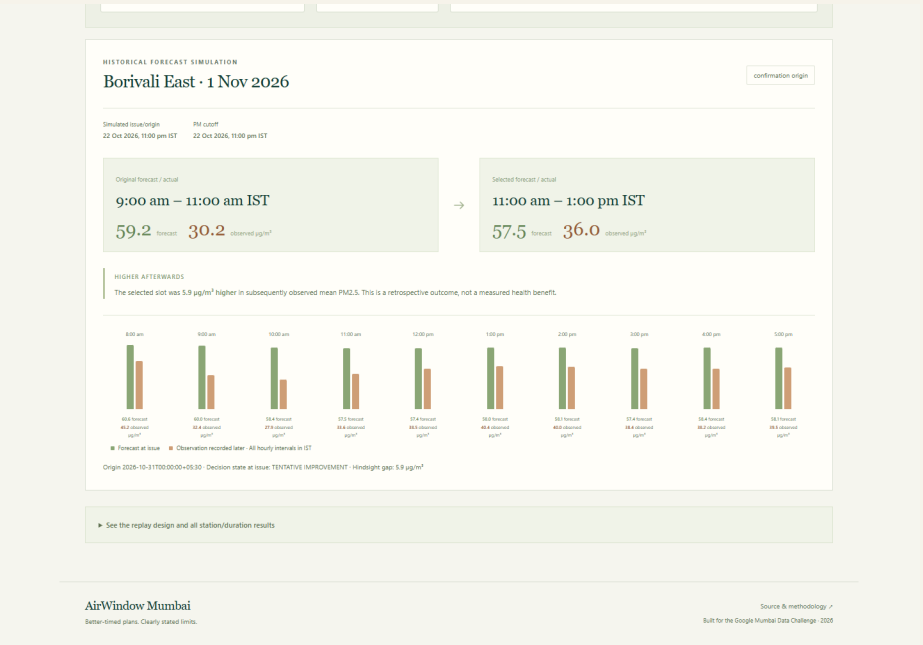
Why this is distinctive

IITM/SAFAR, Google Air View+, Google Air Quality API, London alerts and AirQo already provide forecasts, local information or decision support. AirWindow's narrower contribution is the original-event constraint, paired historical decision replay and honest abstention. This is a product hypothesis, not a first-in-Mumbai claim.

The four supported stations are Borivali East, Malad West, Byculla and Shivaji Nagar. A monitor may not represent nearby streets. No route, heatmap, medical, safe-window or personal-exposure claim is made.

02 / HISTORICAL DECISION REPLAY

Show the forecast.
Then show what happened.



Real application replay. This displayed numerical choice was worse afterwards. Subsequent observations are retrospective outcomes, kept separate from the live artifact; the table below includes all declared cases.

All declared cases	1,848 cases across 77 forecast origins
Observed coverage	1,469 paired original/selected outcomes; 379 missing outcomes retained
Numerical alternatives	1,291 lower-forecast alternatives with paired observations
Lower afterwards	63.67% of those numerical alternatives had lower later observed PM2.5
Realised difference	Median selected minus original: -1.17 µg/m³ for numerical alternatives
Hindsight gap	Median 2.06 µg/m³ when every feasible observed comparison is complete (1,221 cases); 248 paired cases unavailable

These are descriptive retrospective outcomes, not a probability for today's event, a measured health benefit or a calibrated recommendation. Calibrated recommendations: zero.

03 / REPRODUCIBLE EVIDENCE

The evidence chain is inspectable.

The declared scenario uses a 09:00 original booking, 08:00-18:00 feasible range, durations 1/2/3 hours, both target days, all four stations and every matched OOF origin. Numerical selection occurs before inspecting outcomes. Missing observations remain in coverage.

Forecasts are genuine retrospective rolling-origin backtests with a frozen PM cutoff. They were generated on 8 November, using simulated historical origins and an interval-end availability proxy. They are not archived operational first-publication vintages.

Underlying P_fix full-target raw-scale RMSE: 757.9880 on 11,001 development rows, 15.7963 on 1,400 later rows. Eight extreme labels drive 99.846% of development squared error; later rows have no such extremes. These populations are not interchangeable.

Chronological split	69 development origins; 8 later confirmation origins. Cases overlap within origins and are not independent samples.
Later-origin diagnostic	192 declared cases, 180 paired outcomes, 158 numerical alternatives; 60.76% lower afterwards.
Dependence	Exploratory resampling uses contiguous blocks of four forecast origins, not independent hourly errors.
Calibration limit	Later historical origins had prior analyst exposure. No independent calibration, live probability or uncertainty interval is claimed.
Lead limit	OOF issue equals PM cutoff at 193-240h lead. The current live issue is later than the PM cutoff; matching PM age does not establish matching issue-lead calibration.

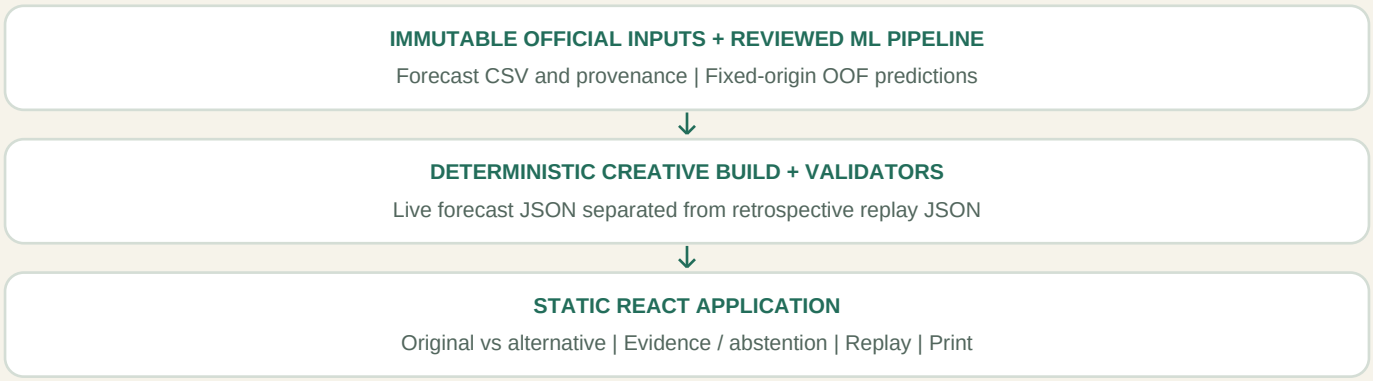
Current forecast provenance

Model: P_fix / TASK008 provisional. Issue: 2026-11-08T15:58:00+05:30. PM-history cutoff and latest station observation: 2026-11-02T23:00:00+05:30. No weather was used. Final official refresh remains pending.

Training SHA-256: fcee20ebf7bd9c4a4e2487bab0a4f81339d6c820a53862b14443031db1f18ad6
Prediction SHA-256: 3d79cffcca78cea1ad6f5e24d2568bf931bbba8c3b15fcd7db1728462050ab1d

04 / ARCHITECTURE, TOOLS AND HANDOFF

A stable demo, without a live API.



Google tools: Firebase Hosting Spark configuration and a Colab-compatible reproducibility notebook. The notebook is not a live application dependency.

Deployment status

The application runs locally and static deployment configuration is prepared. Firebase deployment is pending authenticated CLI/project access. No billing was enabled, and no public hosting URL or cloud-notebook execution is claimed.

Working source and reproducibility

Source: github.com/acchasujal/aqi - existing feature branch
Run/build instructions, tests and notebook: apps/airwindow/
Method, red-team review, real screenshots and submission checklist: reports/creative_submission/

Selected primary references

IITM SAFAR project · Google Air Quality API · AirQo AI Platform
London City Hall alerts · Breathe London · Firebase Hosting quotas

Data and use limits

Uses official challenge data and genuine model artifacts; no extra environmental dataset is required. Four stations do not support street/building predictions. No user interviews, measured health benefit, award guarantee, deployment or form-processing result is invented. Consult official CPCB/government guidance for health decisions.

Creative deadline: 13 November 2026, 18:29 GMT / 23:59 IST. Recheck the official form and submit through the human entrant; this package is not submission confirmation.